

Correlation & Distance

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```
library(bruceR)
library(rstatix)
library(tidyverse)
library(stargazer)
```

1 Correlation

1.1 Bivariate Correlation

Here we're to discuss more about bivariate correlation when taking groups into account. As you can see, the pooled data may interact and neutralize

the effect or relationship of two variables. However, after we inspect that correlation under each group, surprising outcomes may spring up!

Take dataset “car_sales.sav” as an example, which you have been quite similar to “mtcars”. We’re interested in whether or not people value fuel-efficient cars more.

```
carsales = import("car_sales.sav")
```

```
## Successfully imported: 157 obs. of 26 variables
```

```
attach(carsales)
```

```
ggplot(data = carsales, mapping = aes(x = mpg, y = sales)) +  
  geom_point()
```

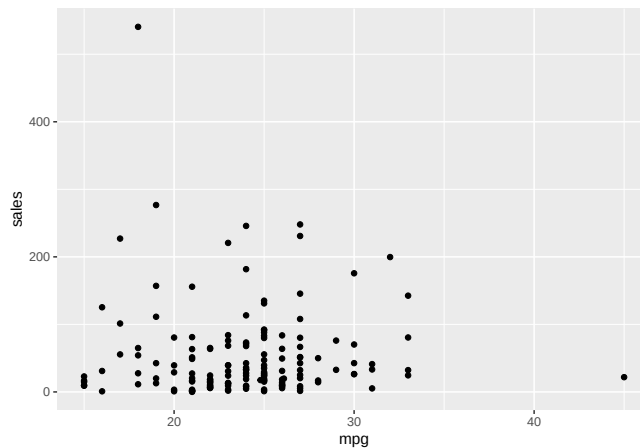


图 1: Scatter Plot of mpg and sales with ALL Observations

At first glance, there are apparently some outliers. Moreover, the scatter points seem to cluster in the bottom-left, possibly a skewed distribution instead of normal! Nevertheless, those don’t bother us to check the overall correlation first.

```
cor(sales,mpg,use = 'c')
```

```
## [1] -0.01668058
```

The correlation is not high in magnitude. Here, use “`cor()`” for simplicity. Note that for data with missing values, specify parameter “`use = 'complete.obs'`” (or “`use = 'c'`” as abbreviation) to compute only with complete cases.

Admittedly, “`cor.test()`” will also return the correlation coefficient, along with outcomes with hypothesis testing. However, the latter makes it to be complex unnecessarily! (Good thing is that, “`cor.test()`” will automatically wipe out cases with missing values.)

```
cor.test(mpg,sales)
```

Then, we’re to deal with outliers and skewness, then to recompute correlation.

Good news is that, log transformation helps when it comes to skewed distribution, and outliers will get contained under log transformation at the same time.

```
cor(log(mpg),log(sales),use = 'c')
```

```
## [1] 0.1141838
```

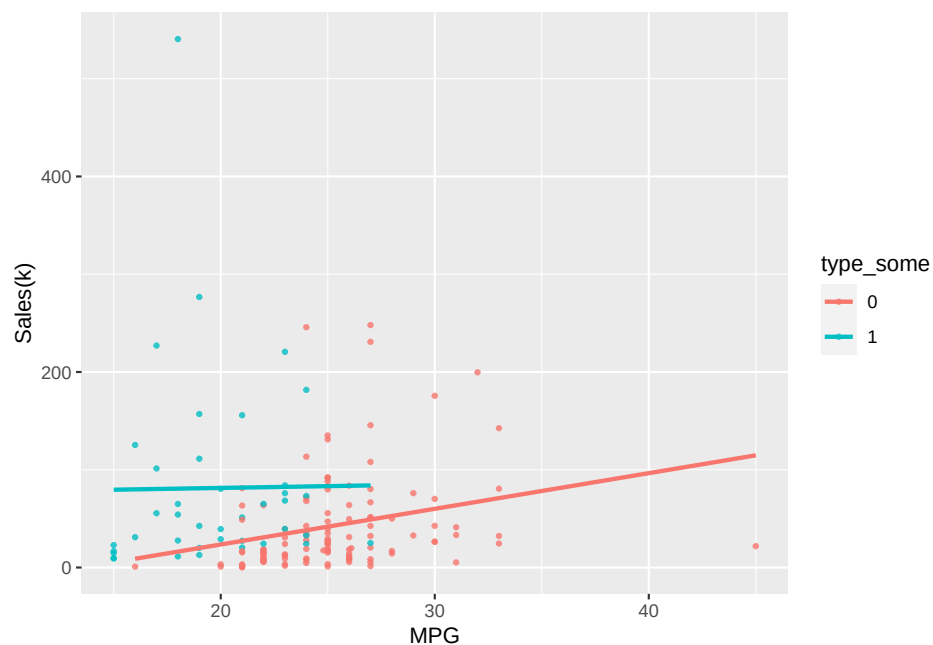
The correlation coefficient is improved in magnitude, but not significant after all.

Then, here comes what’s the most important point. When faced with a brand-new data, before conducting any computation and hypothesis testing, getting an overall picture of the data is the most important thing to do!

After observation, it’s likely that cars whose model are “Metro” or “F-Series” are special compared to others. Accordingly, we’re to group the data and then recompute correlation.

```
sales_some = sales[model!="Metro"&model!="F-Series"]  
mpg_some = mpg[model!="Metro"&model!="F-Series"]  
type_some = as.factor(type[model!="Metro"&model!="F-Series"])
```

```
df = data.frame(sales_some,mpg_some,type_some)
df %>%
  ggplot(aes(mpg_some, sales_some,color = type_some)) +
  geom_point(alpha=0.8, size=0.8) +
  geom_smooth(method="lm",se=FALSE) +
  ylab('Sales(k)') +
  xlab('MPG')
```



Use graphs to think! The categorized data plays an important role here, and unfolds the underlying correlation. Based on the prior observation, then check correlation quantitatively.

```
cor.test(sales_some[type_some=="1"],mpg_some[type_some=="1"])

##
## Pearson's product-moment correlation
##
## data: sales_some[type_some == "1"] and mpg_some[type_some == "1"]
## t = 0.070377, df = 38, p-value = 0.9443
```

```
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.3011645 0.3217809
## sample estimates:
##      cor
## 0.01141584

cor.test(sales_some[type_some=="0"],mpg_some[type_some=="0"])

##
## Pearson's product-moment correlation
##
## data:  sales_some[type_some == "0"] and mpg_some[type_some == "0"]
## t = 2.8695, df = 112, p-value = 0.004915
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.0817139 0.4251484
## sample estimates:
##      cor
## 0.2616958
```

Finally, don't forget to

```
detach(carsales)
```

1.2 Partial & Part Correlation

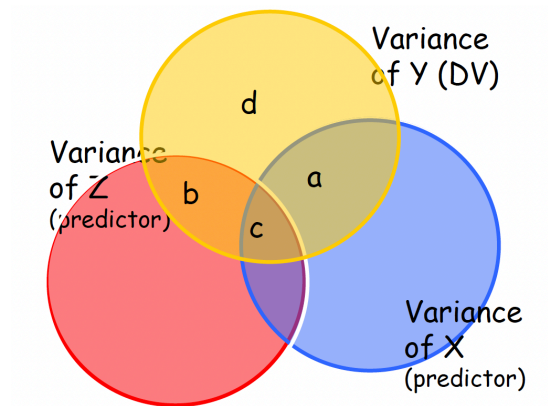
1.2.1 Background Knowledge

Above all, I'd like to clarify one concept that we've been familiar with but without mentioning this concept. Zero-order correlation refers to the correlation between X and Y without controlling other covariates. Traditionally, Pearson and Spearman correlations are typical of zero-order correlation.

We've discussed partial and part (or, semi-partial) correlation in multilinear

regression. Take a step further, we can consider those under the most general cases, i.e., without the setting of MLR but some variables.

Suppose we have three variables, called X, Y and Z , where X is the predictor, Y is the dependent variable, and Z is the covariate. Noteworthy that the assumption of the roles of X, Y, Z don't matter much, only for accommodating the setting.



Note that, in the general case, Y is not represented as the whole area in the picture, but just an equal role as a circle.

If the correlation of X and Y is of our great interest,

$$r_x = \frac{a + c}{a + b + c + d}$$

$$pr_x = \frac{a}{a + d}$$

$$spr_x = \frac{a}{a + b + c + d}$$

- Partial: Holding Z from BOTH X and Y , i.e., removing the effect of Z .

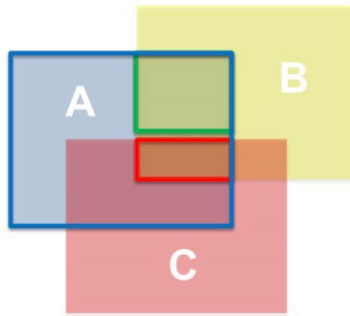
Good for knowing the **variance uncounted by other control variable(s)**.

- Semi-Partial: Holding Z from X only.

Good for knowing the **unique contribution of X** .

- Zero-Order: No extra constraint.

One more enlightening way to think about partial correlation is through multilinear regression.



If we're interested in the partial correlation between A and B , with C as the covariate. Then, what we need to do is partial out the effect of C on BOTH A and B , and this goal can be accomplished through step-by-step linear regression.

Specifically, first we conduct a simple linear regression of A on C , and the residual part is that of A which doesn't intersect with C . Similarly, then followed by a SLR of B on C , same with the meaning of residual part. For the two residual parts, they share it in common that they're clean of C . Lastly, compute the correlation between A and B directly, and that's what we want!

Even with more covariates, conduct A and B on all the covariates and then compute the correlation with their residual parts. Done!

1.2.2 Examples

We first man-make a data that vividly shows the differences between partial and zero-order correlation.

```
DV = c(1:8, 8:11)
IV = c(4:1, 8:5, 12:9)
```

```

covariate = c(rep(0,4), rep(4,4), rep(7,4))
studiedData = data.frame(DV,IV,covariate)

studiedData |> ggplot(aes(x = IV,y = DV))+
  geom_point()

```

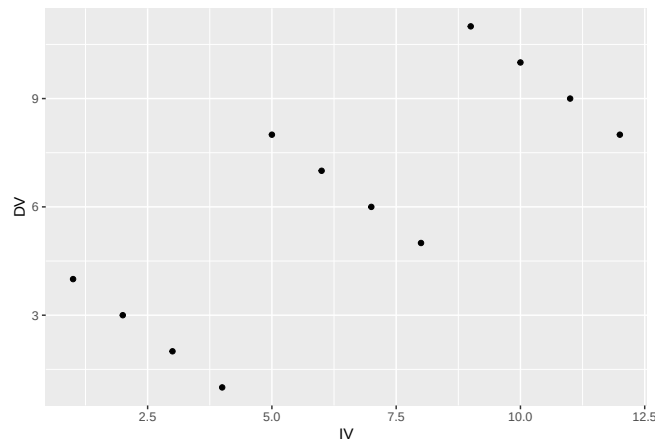


图 2: Scatter Plot of X and Y

Clearly, considering zero-order correlation, X is strictly negatively-correlated with Y , the covariate is strictly positively-correlated with Y . However, X shows partially positive correlation with Y overall.

```
cor(IV,DV)
```

```
## [1] 0.7608292
```

If you're to study the partial correlation, things will be different.

```

library(ppcor)
pcor(studiedData)

```

One more example. A popular radio talk show host has just received the latest government study on public health care funding and has uncovered a startling fact: As health care funding increases, disease rates also increase! Cities that spend more actually seem to be worse off than cities that spend

表 1: Partial Correlation between Variables

	DV	IV	covariate
DV	1	-0.972	0.991
IV	-0.972	1	0.993
covariate	0.991	0.993	1

less. Then, shall we just cut the funding and save people's health?

```
# example: a covariate can "make" a correlation that should not exist
currentData <- import("health_funding.sav")
```

```
## Successfully imported: 50 obs. of 4 variables
```

```
attach(currentData)
studiedData = data.frame(disease,funding,visits)
```

First of all, check overall zero-order correlation.

```
cor(studiedData)
```

表 2: Zero-Order Correlation

	disease	funding	visits
disease	1	0.737	0.762
funding	0.737	1	0.964
visits	0.762	0.964	1

It seems astonishing that “**funding**” is greatly correlated with “**disease**”! May it offer the plausible reason to cut the funding? Remember, we still have the covariate “**visits**”. One alternative explanation might be that, offered with more funding, people may be more likely to pay visits to health care, and thus more diseases are detected, which should have been detected

but not due to wealth state.

```
pcor(studiedData)$estimate
```

表 3: Partial Correlation

	disease	funding	visits
disease	1	0.013	0.286
funding	0.013	1	0.920
visits	0.286	0.920	1

It's astonishing that, considering the partial correlation, “**funding**” almost has no correlation with “**disease**” in essence!

Additionally, we can consider the part correlation.

```
spcor(studiedData)$estimate
```

表 4: Part Correlation

	disease	funding	visits
disease	1	0.009	0.193
funding	0.004	1	0.622
visits	0.076	0.596	1

And finally,

```
detach(currentData)
```

2 Distance

We need to distinguish between two kinds of distance. One is distance between cases (or observations), the other is distance between variables.

There are traditionally three ways to measure distance, either from the perspective of dissimilarity or similarity.

- Dissimilarity
 - Euclidean distance

$$d(\vec{p}, \vec{q}) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

- Chebychev distance

A vector space where the distance between two vectors is the greatest of their differences along any coordinate dimension.

$$D(\vec{x}, \vec{y}) = \max_i |x_i - y_i|$$

- Similarity
 - Cosine similarity
 - * A measure of similarity between two non-zero vectors of an inner product space that measures the cosine of the angle between them.
 - * It is thus a judgment of orientation and not magnitude.

```
currentData = import("data10-04.sav", sheet = 1)
```

```
## Successfully imported: 12 obs. of 6 variables
```

```
attach(currentData)
```

```
# only use the data that should be considered or analyzed
```

```
studiedData = data.frame(hgrow,temp,rain,hsun,humi)
```

2.1 Correlation

Note that correlation itself is a measure of distance. However, correlation only applies for **distance between variables**, not for distance between cases. (If you insist to test the correlation between cases, easy, just transpose your data!)

```
cor(studiedData)
```

表 5: Correlation between Variables

	hgrow	temp	rain	hsun	humi
hgrow	1	0.983	0.709	0.704	0.374
temp	0.983	1	0.715	0.690	0.292
rain	0.709	0.715	1	0.702	0.384
hsun	0.704	0.690	0.702	1	-0.051
humi	0.374	0.292	0.384	-0.051	1

2.2 Euclidean Distance

Then, we're interested in Euclidean distance **between cases** by default. Use “`dist(x, method = "euclidean")`”. Mind that the distance makes sense only if the data is scaled!

```
# studiedData %>% dist()
studiedData %>%
  scale() %>% dist()
```

If the Euclidean distance between **variables** is of interest, then *transpose* your data.

```
dist = studiedData %>%
  scale() %>% t() %>% dist()
```

It's noteworthy that, the Euclidean distance result returned by “`dist()`” is a unique type of “`dist`”, which is incompatible with some other types. Luckily, use “`as.matrix()`” to transform that into a matrix (with the diagonal and upper right triangle filled automatically).

表 6: Euclidean Distance between Cases

	1	2	3	4	5	6	7	8	9	10	11
1	0	0.925	2.142	3.692	4.790	3.904	4.078	5.793	3.764	2.536	1.782
2	0.925	0	1.288	2.856	4.179	3.338	3.756	5.505	3.576	2.232	1.433
3	2.142	1.288	0	2.149	3.774	3.151	3.962	5.710	3.870	2.675	2.054
4	3.692	2.856	2.149	0	2.558	2.001	3.108	4.535	3.518	2.627	2.509
5	4.790	4.179	3.774	2.558	0	2.307	2.871	2.895	2.724	3.151	3.662
6	3.904	3.338	3.151	2.001	2.307	0	1.265	3.338	1.937	1.636	2.236
7	4.078	3.756	3.962	3.108	2.871	1.265	0	2.696	1.292	1.566	2.448
8	5.793	5.505	5.710	4.535	2.895	3.338	2.696	0	2.761	3.721	4.519
9	3.764	3.576	3.870	3.518	2.724	1.937	1.292	2.761	0	1.654	2.541
10	2.536	2.232	2.675	2.627	3.151	1.636	1.566	3.721	1.654	0	0.967
11	1.782	1.433	2.054	2.509	3.662	2.236	2.448	4.519	2.541	0.967	0
12	0.591	0.923	2.192	3.447	4.501	3.529	3.629	5.333	3.402	2.093	1.374

表 7: Euclidean Distance between Variables

	hgrow	temp	rain	hsun	humi
hgrow	0	0.605	2.529	2.550	3.712
temp	0.605	0	2.505	2.609	3.947
rain	2.529	2.505	0	2.561	3.680
hsun	2.550	2.609	2.561	0	4.808
humi	3.712	3.947	3.680	4.808	0

2.3 Cosine Similarity

Also, we can see the cosine similarity **between cases**.

Remember, if you're to study the distance between cases, transpose your matrix of data; if you're to study the distance between variables, throw the matrix of data into “`lsa::cosine()`” directly.

```
studiedData %>%
  as.matrix.data.frame() %>% t() %>%
  lsa::cosine() %>%
  stargazer(title = 'Cosine Similarity between Cases')
```

表 8: Cosine Similarity between Cases

1	0.985	0.968	0.886	0.626	0.886	0.851	0.632	0.643	0.934	0.970	0.994
0.985	1	0.994	0.948	0.609	0.901	0.846	0.616	0.597	0.931	0.995	0.997
0.968	0.994	1	0.974	0.664	0.934	0.881	0.671	0.639	0.950	0.990	0.986
0.886	0.948	0.974	1	0.662	0.927	0.860	0.669	0.600	0.913	0.954	0.925
0.626	0.609	0.664	0.662	1	0.876	0.929	1.000	0.984	0.847	0.571	0.609
0.886	0.901	0.934	0.927	0.876	1	0.989	0.882	0.852	0.991	0.890	0.894
0.851	0.846	0.881	0.860	0.929	0.989	1	0.934	0.920	0.982	0.828	0.847
0.632	0.616	0.671	0.669	1.000	0.882	0.934	1	0.984	0.853	0.580	0.616
0.643	0.597	0.639	0.600	0.984	0.852	0.920	0.984	1	0.843	0.556	0.610
0.934	0.931	0.950	0.913	0.847	0.991	0.982	0.853	0.843	1	0.916	0.932
0.970	0.995	0.990	0.954	0.571	0.890	0.828	0.580	0.556	0.916	1	0.990
0.994	0.997	0.986	0.925	0.609	0.894	0.847	0.616	0.610	0.932	0.990	1

And finally,

```
detach(currentData)
```